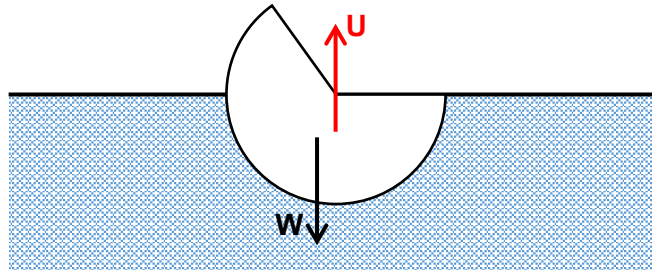


Tutorial 2 Forces and Moments Suggested Solutions

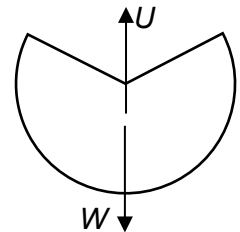
D1 (a)



Note: Weight acts from the centre of gravity of the object. Upthrust acts from the centre of gravity of the displaced fluid.

- (b)** W and U will exert an anticlockwise torque on the object causing it to rotate anticlockwise. It will then oscillate until it settles down to its equilibrium position.

At the equilibrium position, the lines of action of both W and U are aligned (i.e. no perpendicular distance between them).



- D2 (a) (i)** For $T_2 = 0$, T_1 must be parallel to the weight of the object and supporting the entire weight of the object.

Hence, $T_1 = 60.0 \text{ N}$ and $\theta = 90^\circ$.

- (ii)** In the horizontal direction,

$$T_1 \cos \theta = 70.0 \cos 50^\circ$$

$$T_1 \cos \theta = 44.995 \text{ ----- (1)}$$

In the vertical direction,

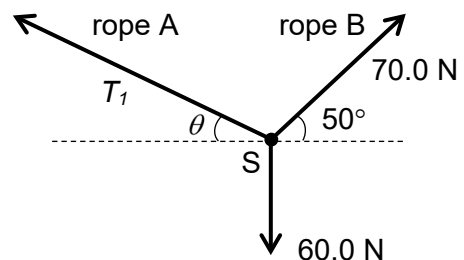
$$T_1 \sin \theta + 70.0 \sin 50^\circ = 60.0$$

$$T_1 \sin \theta = 60.0 - 70.0 \sin 50^\circ = 6.3769 \text{ ----- (2)}$$

$$\frac{(2)}{(1)}: \tan \theta = 0.14172$$

$$\theta = 8.066 = 8.07^\circ$$

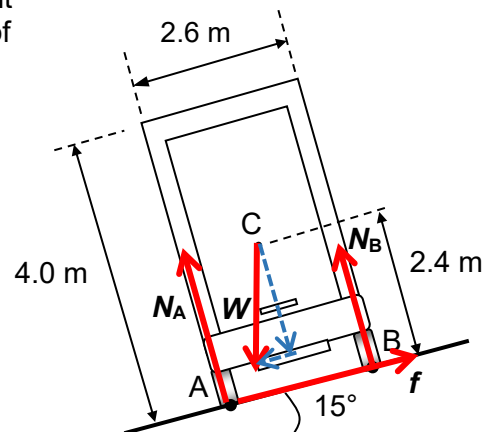
$$\text{from (1): } T_1 = \frac{44.995}{\cos 8.07^\circ} = 45.445 = 45.4 \text{ N}$$



- (b)** No, it would not be possible. If both ropes A and B are completely horizontal, there would be no vertical component of their tensions to support the weight of the object, which is acting vertically downwards. Hence, the object cannot be in equilibrium.

- D3 (a)** Since the frictional force f acts along the slope, it must be equal in magnitude to the component of the weight along the slope in order for the truck to be in equilibrium.

$$\begin{aligned} f &= W \sin 15^\circ \\ &= (300)(9.81) \sin 15^\circ \\ &= 761.70 \text{ N} = 762 \text{ N} \end{aligned}$$



- (b)** N_A is larger than N_B .

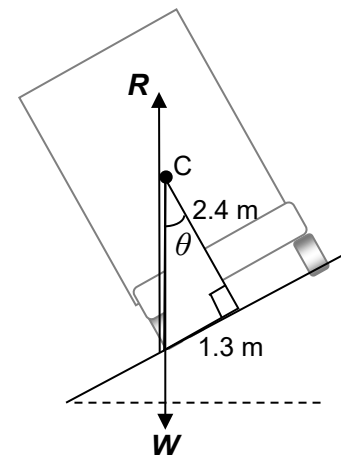
Taking moments about point C, both N_B and f contribute to the anti-clockwise moments, while only N_A contributes to the clockwise moments.

Hence, for the truck to be in rotational equilibrium (with zero net moment about C), N_A would need to have a greater magnitude than N_B .

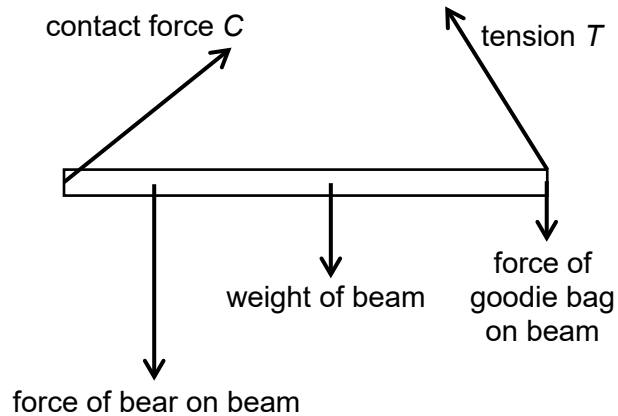
- (c)** (i) 0 N. When the truck is about to topple over, it will pivot about point A and point B will lose contact with the slope. Hence, $N_B = 0 \text{ N}$.
- (ii) When the truck is about to topple over, W must be acting directly through point A (the pivot point).

In other words, at the critical angle, the line of force of the weight of the truck coincides with the contact force at the left wheel. Hence,

$$\begin{aligned} \tan \theta &= \frac{1.3}{2.4} \\ \theta &= 28.443 = 28.4^\circ \end{aligned}$$



D4 (a)



(b) Taking moments about the hinge,

$$(700)(1.0) + (200)(3.0) + (80)(6.0) = (T \sin 60^\circ)(6.0)$$

$$T = 342.56 = 343 \text{ N}$$

Consider forces in the horizontal direction,

$$C_x = T \cos 60^\circ = 342.56 \cos 60^\circ = 171.28 \text{ N}$$

Consider forces in the vertical direction,

$$C_y + T \sin 60^\circ = 700 + 200 + 80$$

$$C_y = 980 - 342.56 \sin 60^\circ = 683.33 \text{ N}$$

$$C = \sqrt{C_x^2 + C_y^2} = \sqrt{171.28^2 + 683.33^2} = 704.47 \text{ N} = 704 \text{ N}$$

angle θ anticlockwise above the horizontal

$$= \tan^{-1} \left(\frac{C_y}{C_x} \right)$$

$$= \tan^{-1} \left(\frac{683.33}{171.28} \right)$$

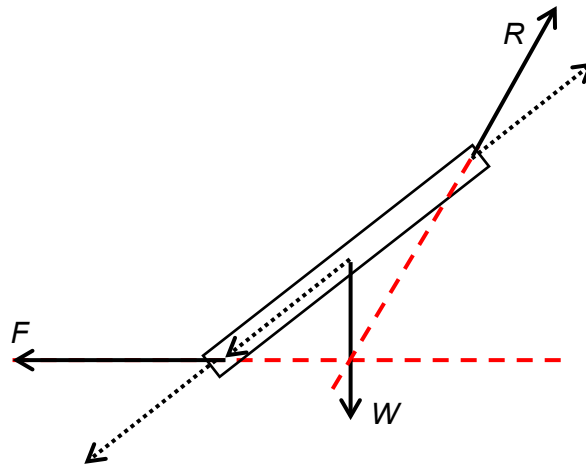
$$= 75.928^\circ = 75.9^\circ$$

(c) Taking moments about the hinge,

$$700x + 200(3.0) + 80(6.0) = (900 \sin 60^\circ)(6.0)$$

$$x = 5.1379 = 5.14 \text{ m}$$

D5 (a)



The plank is experiencing tension (i.e. it is being “stretched”).

The spring force F and weight W have components along the plank downslope, while the contact force R has a component along the plank upslope. Since the downslope and upslope components are acting outwards along the plank, (indicated by the dotted arrows), these forces act to “stretch” the plank. Hence the plank is in tension.

(b) Taking moments about the hinge,

$$W \left(\frac{1}{2} L \sin 60^\circ \right) = F (L \cos 60^\circ)$$

$$F = \frac{W}{2} \tan 60^\circ = \frac{12.0 \times 9.81}{2} \tan 60^\circ = 101.9 \text{ N} = 102 \text{ N}$$

(c) By Newton's 3rd Law, $F_{\text{spring on plank}} = F_{\text{plank on spring}} = kx$

$$101.9 = (150) x$$

$$x = 0.67933 = 0.679 \text{ m}$$

(d) $R = \sqrt{W^2 + F^2}$

$$= \sqrt{(12.0 \times 9.81)^2 + (101.9)^2}$$

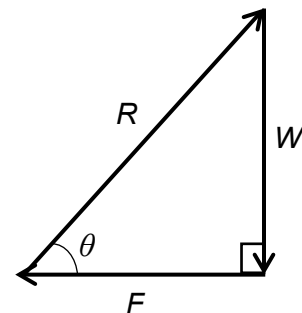
$$= 155.697 \text{ N} = 156 \text{ N}$$

angle θ anticlockwise above the horizontal

$$= \tan^{-1} \left(\frac{W}{F} \right)$$

$$= \tan^{-1} \left(\frac{12.0 \times 9.81}{101.9} \right)$$

$$= 49.120^\circ = 49.1^\circ$$



- D6 (a)** Let x be the distance of the c.g. from the bottom of the mallet

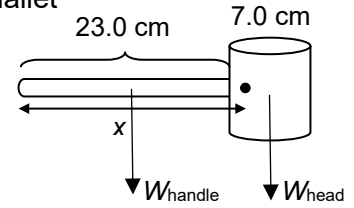
Taking moments about the c.g.,

$$W_{\text{handle}} x_{\text{handle-cg}} = W_{\text{head}} x_{\text{head-cg}}$$

$$0.45(9.81) \times \left(x - \frac{23.0}{2} \right) = 1.80(9.81) \times \left((23.0 + \frac{7.0}{2}) - x \right)$$

$$4.4145x - 50.76675 = 467.937 - 17.658x$$

$$x = 23.5 \text{ cm}$$



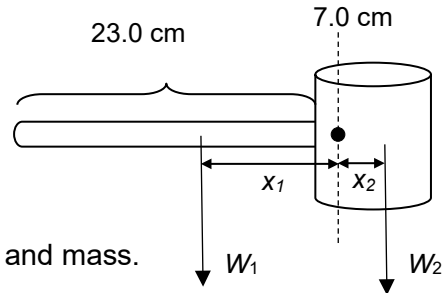
- (b)** No, the student will not be successful.

The net moment about the c.g. is zero.

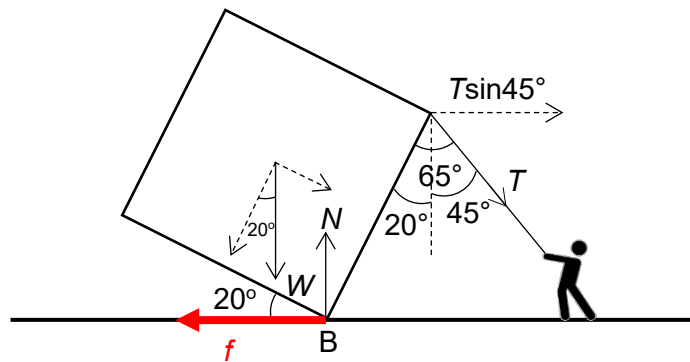
$$\text{Hence, } W_1 x_1 = W_2 x_2$$

But since $x_1 > x_2$, $W_1 < W_2$.

Hence the two pieces will not be of equal weight and mass.



- D7 (a)**



Taking moments about B,

$$\begin{aligned} \text{Moment due to } W &= (W \cos 20^\circ) \left(\frac{L}{2} \right) - (W \sin 20^\circ) \left(\frac{L}{2} \right) \\ &= (50)(9.81) \left(\frac{3.0}{2} \right) (\cos 20^\circ - \sin 20^\circ) \\ &= 439.74 = 440 \text{ N m (anticlockwise)} \end{aligned}$$

- (b)** By principle of moments, taking moments about B
anticlockwise moment = clockwise moment

$$440 = (T \sin 65^\circ)(3.0)$$

$$T = 161.83 = 162 \text{ N}$$

- (c) (i) Frictional force f is pointing to the left, parallel to the floor.

The tension T has a horizontal component acting to the right. In order to maintain translational equilibrium, there needs to be a force acting to the left.

Since the weight and normal contact force are both vertical, the only force that can oppose the tension's horizontal component is the frictional force.

Hence, frictional force must be acting to the left.

- (ii) To maintain translational equilibrium,
 $f = T \sin 45^\circ$
 $= 162 \sin 45^\circ$
 $= 114.55 = 115 \text{ N}$

C1 Ans: E

Method 1

By balancing the forces at the connection points.

Starting from rods P and R. Since W is downwards, R must be diagonally upwards to the right and P must be to the left. Hence R is in compression and P in tension.

Looking at the connection between Q, R and S. Since R is in compression, the force is diagonally downwards to the left. Q must be diagonally upwards to the left to balance the downward force and S must be to the right to balance the left forces from Q and R. Hence Q is in tension and S in compression.

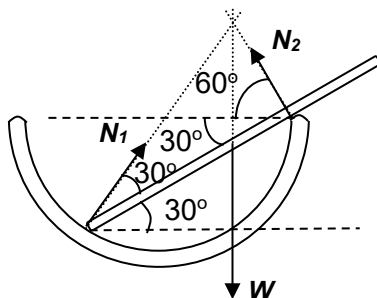
Method 2

Imagine all the rods as springs.

When you have a downward force W on spring P and R, P will stretch and R will compress. This will in turn compress S and stretches Q. Hence R, S are in compression and P, Q are in tension.

***Although this method is faster and easier to grasp, it doesn't really help in understanding your concepts.*

C2



Horizontally,

$$N_2 \cos 60^\circ = N_1 \cos 60^\circ$$

$$\Rightarrow N_2 = N_1$$

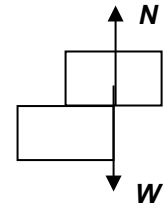
Vertically,

$$N_2 \sin 60^\circ + N_1 \sin 60^\circ = W$$

Solving,

$$N_1 = N_2 = 0.58 W \quad \left(\text{i.e. } \frac{W}{\sqrt{3}} \right)$$

C3 (a) (i) Extension = $\frac{L}{2}$

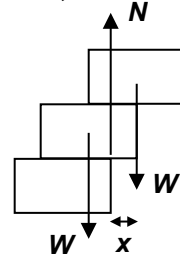


- (ii)** Considering the top 2 blocks as one object acting on the bottom block.
Taking moment about the top right corner of the bottom block,

$$Wx = W\left(\frac{L}{2} - x\right)$$

$$x = \frac{L}{4}$$

$$\text{Extension} = \frac{L}{4}$$

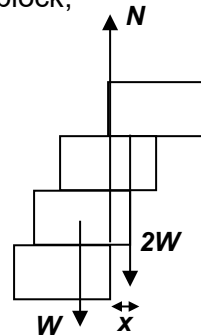


- (iii)** Considering the top 3 blocks as one object acting on the bottom block.
Taking moment about the top right corner of the bottom block,

$$2Wx = W\left(\frac{L}{2} - x\right)$$

$$x = \frac{L}{6}$$

$$\text{Extension} = \frac{L}{6}$$



- (b)** Consider n number of blocks on the bottom block,
Taking moment about the top right corner of the bottom block,

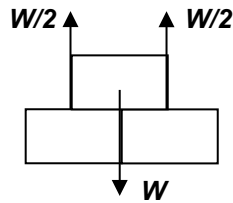
$$(n-1)Wx = W\left(\frac{L}{2} - x\right)$$

$$x = \frac{L}{2n}$$

Hence if total extension for n number of blocks on the bottom block

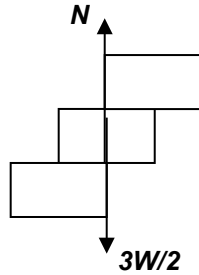
$$= \sum_{i=1}^n \frac{L}{2i}$$

(c)

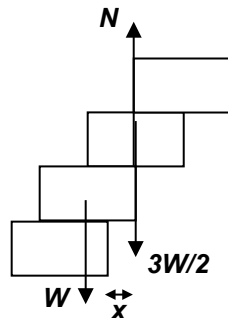


Forces acting on the top block

If both blocks at the bottom are about to topple, the normal reaction forces acting on the top block would be at the two furthest sides.



Not counting the top block, extension of 1st block = $L/2$



Taking moment about the top right corner of the bottom block,

$$\frac{3}{2}Wx = W\left(\frac{L}{2} - x\right)$$

$$x = \frac{L}{5}$$

Extension of 2nd block = $L/5$

Similarly it can be found that the extensions for the 3rd, 4th, n blocks are $L/7$, $L/9$,

$\frac{L}{2(n-1)+3}$ respectively.

Hence total distance spanned by n blocks = $\sum_{i=1}^n \frac{L}{2(i-1)+3} + \frac{L}{2}$

$$\therefore 2\left(\sum_{i=1}^n \frac{L}{2(i-1)+3} + \frac{L}{2}\right) = 1.0 \text{ m}$$

Using Excel Spreadsheet, $n = 40$.

Therefore total number of blocks needed (including the bottom 2 blocks that are not overhanging and the top block) = $40 + 3 = 43$.